



PYTHON PROJECT ASSIGNMENT

For Python with AI Students

Project Title	Student Performance Analytics System
Technology	Python, NumPy, Pandas
Concepts Covered	Variables, Data Types, Loops, Functions, NumPy, Pandas, Data Analysis
Submission Deadline	15 October 2026 (on or before)
Final Submission	Project + GitHub Repository + Documentation + Google Form Link Submission

Project Goal

Build a Python-based Student Performance Analytics System that reads student data, processes and analyzes it using NumPy and Pandas, and produces meaningful results such as marks summary, averages, grades, pass/fail statistics, subject-wise performance, and top performers.

Important: This is a Python data-analysis project. Students are expected to use the concepts already covered in class. Advanced machine learning is not required for this assignment.

Required Project Features

- Create or use a student dataset containing fields such as Student ID, Name, Department, Subjectmarks and Attendance.
- Load and organize the data using Pandas DataFrame.
- Use NumPy for numerical calculations wherever appropriate.
- Create reusable functions for calculations and processing.
- Use loops where iteration is required.
- Calculate total marks and average marks for each student.
- Assign grades based on a clearly defined grading rule.
- Identify pass/fail students using a clearly defined condition.
- Find highest, lowest and average performance.
- Generate subject-wise performance analysis.
- Identify top-performing students.

1. Suggested Project Structure

Students may organize the project in a simple and readable structure. The exact file names may vary, but the repository should be easy for another person to understand.

Item	Suggested Name	Purpose
Main program	main.py	Runs the project and displays the final analysis
Data	students.csv	Stores student records
Functions	functions.py (optional)	Reusable calculations/processing functions
Documentation	Project_Documentation	Explains project, implementation and outcome

2. Implementation Guidelines

Step 1 – Prepare the Dataset

Create a CSV file with at least 15–20 student records. Include realistic values for student details, marks and attendance. Avoid using only one or two records.

Step 2 – Read the Data

Use Pandas to read the CSV file and display the data. Check the columns, number of records and basic information.

Step 3 – Process the Data

Use functions, loops, conditions, NumPy and Pandas operations to calculate totals, averages, grades, pass/fail status and other required metrics.

Step 4 – Perform Analysis

Answer practical questions from the dataset, such as: Which student has the highest average? Which subject has the best average? How many students passed? What is the overall class average?

Step 5 – Final Output

Present the results clearly. The output should be understandable without needing to inspect the code.

3. Student Rules & Guidelines

- The project must be completed individually unless the trainer explicitly assigns teams.
- Use Python concepts already taught in the course: functions, loops, NumPy and Pandas.
- Do not simply copy a complete project from the internet or another student.
- Students may refer to documentation/tutorials for learning, but the final implementation should be their own work.
- Use meaningful variable names, function names and comments where necessary.
- Keep the project files organized and remove unnecessary files before pushing to GitHub.
- Test the project before submission and make sure the main program runs without avoidable errors.
- The project must be submitted on or before 15 October 2026.

4. GitHub Submission Instructions

After completing and testing the project, students must create a GitHub repository and push the complete project.

1. Create/sign in to your GitHub account.
2. Create a new repository with a clear name, for example: student-performance-analytics.
3. Add the Python files, dataset and documentation files required for the project.
4. Add a README.md file containing the project documentation.
5. Commit and push the complete project to the repository.
6. Open the repository in the browser and verify that all required files are visible.
7. Copy the repository URL.
8. Paste the GitHub repository link into the Google Form provided by EWB Courses.

GitHub Checklist

✓	Repository is public/accessibility is set as instructed by the trainer.
✓	Main Python file is uploaded.
✓	Dataset is uploaded.
✓	README.md is included.
✓	Project runs successfully.
✓	GitHub link is copied correctly into the Google Form.

5. Project Documentation – Assessment Activity

After completing the project and GitHub submission, students must prepare project documentation. The documentation should explain what was built, how it was implemented and what result was obtained.

Section	What the student should write
1. Project Title	Name of the project and student details.
2. Objective	Why the project was created and what problem it solves.
3. Technologies Used	Python, NumPy, Pandas and any other tools actually used.
4. Dataset Description	What data was used, important columns and approximate number of records.
5. Implementation	Step-by-step explanation of data loading, processing, functions, NumPy/Pandas operations and analysis
6. Key Features	List the main functions/features implemented in the project.
7. Output / Screenshots(optional)	Add screenshots showing important outputs/results.
8. Final Outcome	Explain the final result and what the project successfully achieved.

Section	What the student should write
9. Challenges & Learning	Mention problems faced, how they were solved and what was learned.
10. GitHub Repository	Add the final GitHub repository link.

6.Documentation Submission Workflow

Students should follow this sequence. Completing only the code is not the final submission.

Step	Action	Expected Output
1	Complete the Python project	Working project
2	Test and correct errors	Stable final version
3	Create GitHub repository	Repository created
4	Push project files to GitHub	Complete project online
5	Copy GitHub repository URL	Repository link
6	Submit link in EWB Google Form	Form submission completed
7	Prepare project documentation	Documentation file
8	Push documentation to GitHub	Updated repository
9	Verify final repository	Code + dataset + README + documentation

7.Assessment Focus

The project will be assessed based on the student's understanding and implementation, not just the final output.

Area	Focus
Python Fundamentals	Correct use of functions, loops, conditions and data handling
NumPy	Appropriate use of NumPy for numerical operations
Pandas	DataFrame creation/loading, filtering, analysis and useful operations
Project Logic	Correct calculations, conditions and meaningful results
Code Quality	Readable structure, naming, organization and basic comments
GitHub	Complete repository, correct files and working README
Documentation	Clear objectives, implementation, outputs and final outcome

FINAL SUBMISSION CHECKLIST

Before submitting, make sure you have completed all of the following:

- ☐ Project completed and tested
- ☐ Student dataset included
- ☐ NumPy and Pandas used appropriately
- ☐ Functions and loops used where required
- ☐ GitHub repository created
- ☐ Complete project pushed to GitHub

- ☐ README.md added
- ☐ GitHub repository link submitted through the Google Form
- ☐ Documentation completed with Objectives, Implementation and Final Outcome
- ☐ Documentation/screenshots added to the GitHub repository
- ☐ Final GitHub repository checked before the deadline

DEADLINE: 15 OCTOBER 2026